

The Formulas Behind Each Strategy

Six ways to invest ₹10,000 every month in Nifty 50, Gold and a Liquid fund, explained step by step with the maths, a plain-language example for every step and real rupee numbers from the back-test (Feb 2010 – Dec 2019).

How to read this document

1. **Part A** explains the ideas and symbols used everywhere (read this first).
2. **Strategies 1-6** each go through the SIP engine from start to finish. Every step has: what happens in plain words → the formula → what each symbol means → an example with real numbers.
3. Each strategy then shows **one full month in rupees**, and its **scores after 10 years**.
4. The last pages compare all six and show the **flowchart** of the whole process.

The six strategies

#	Strategy	Target split	Instalment split	Rebalance
1	Equity SIP	100% Nifty	pro-rata	never
2	Equal-weight SIP	⅓ each	pro-rata	never
3	60/20/20 SIP	60 / 20 / 20	pro-rata	every 12 months
4	Risk-parity SIP	optimiser: equal risk	smart	drift > 5%
5	Max-Sharpe SIP	optimiser: best return/risk	smart	drift > 5%
6	Optimized SIP	average of 4 and 5	smart	drift > 5%

All numbers in this document are produced by the project's code (`sip/engine.py`); the example months are recomputed step by step and checked against the engine's own output.

Part A The ideas and symbols used everywhere

The three jars

Picture your SIP as **three jars of money**: a Nifty 50 jar (Indian shares), a Gold jar (gold priced in rupees) and a Liquid jar (a fund that earns the 91-day Treasury-bill interest rate). Every month:

1. ₹10,000 arrives and is **split** between the jars;
2. maybe the jars are **rebalanced** (money moved between them);
3. a small **fee** is paid on everything bought or sold;
4. the **market** makes each jar grow or shrink by that asset's return.

This loop is the **SIP engine** (`sip/engine.py`). All six strategies use the same engine. They differ only in **three settings**: the target split, how the instalment is split, and when to rebalance.

The input: monthly returns

From the daily data (`00_raw_data/`) each asset gets one return per month, the growth of ₹1 over that month:

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1$$

$P_{i,t}$ month-end price of asset i in month t (Nifty adds 1.3%/12 for dividends; gold is converted to rupees: \$ price $\times$ ₹ per \$)

$r_{i,t}$ the return of asset i in month t , e.g. $0.0093 = +0.93\%$

Example: Feb 2010: Nifty +0.93%, Gold +3.17%, Liquid +0.31%. A ₹100 Nifty holding became ₹100.93.

Symbols

t month number: Feb 2010 = 0, ... , Dec 2019 = 118

i which jar: Nifty, Gold or Liquid

C_t the instalment paid at the start of month t (₹10,000)

w_i target share of jar i (e.g. $0.60 = 60\%$); the three add up to 1

h_i rupees currently in jar i ("holdings")

x_i rupees of this month's instalment that go into jar i

$\sum_i$ "add up over all jars"

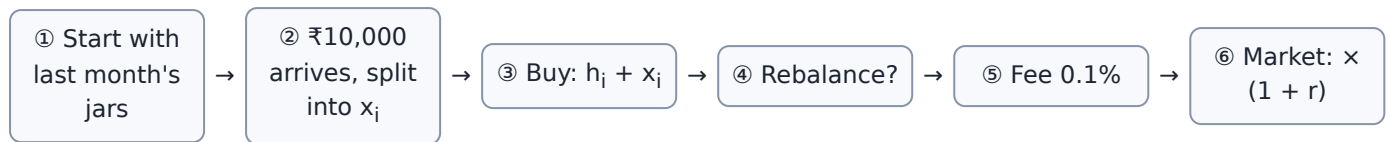
$\leftarrow$ "becomes": the jar's new value replaces the old one

$r_{i,t}$ return of asset i in month t

c trading cost = 0.1% of every rupee bought or sold

μ, Σ average returns and covariance (only used by the optimisers, Strategies 4–6)

The engine in one picture



↻ repeated 119 times (Feb 2010 → Dec 2019), then the strategy is scored.

In each strategy section the steps are numbered 1-9: step 1 sets the target, steps 2-8 are one month of the engine (①-⑥ above, plus measuring the month's return), and step 9 scores the result.

Strategy 1 Equity SIP: 100% Nifty 50

The usual SIP most people do: the whole ₹10,000 goes into a Nifty 50 index fund every month. It is the benchmark the other five are compared with.

Setting	This strategy
Where the target split w comes from	Fixed: 100 / 0 / 0
How each instalment is split	Pro-rata (all of it to Nifty)
When the whole portfolio is rebalanced	Never
Instalment, fee, market move, TWR, scoring	Same for all six strategies

The steps and their formulas

1 Choose the target split (fixed)

Everything goes to Nifty. Gold and Liquid get nothing.

$$w = (w_{Nifty}, w_{Gold}, w_{Liquid}) = (1, 0, 0)$$

w_i target share of jar i ; the three shares add up to 1 (100%)

Example: the same target is used in every one of the 119 months. Nothing is learned from data.

2 The instalment arrives

On the 1st of every month you pay in the same amount. The formula allows the amount to grow every year (a “step-up”), but in this project the growth rate is 0, so it is always ₹10,000.

$$C_t = A \times (1 + g)^{\lfloor t/12 \rfloor}$$

C_t the instalment paid in month t

t the month number, counting from 0 (Feb 2010 is $t = 0$, Dec 2019 is $t = 118$)

A the starting instalment = ₹10,000

g yearly step-up of the instalment = 0 (no step-up)

$\lfloor t/12 \rfloor$ number of full years passed ($t \div 12$, rounded down)

Example: Feb 2010: $C = 10,000 \times (1 + 0)^0 = \text{₹10,000}$. Because $g = 0$, every one of the 119 months is ₹10,000, so the total paid in is $119 \times \text{₹10,000} = \text{₹1,190,000}$.

3 Split the instalment (pro-rata)

The ₹10,000 is cut into pieces in exactly the target proportions, like cutting a cake using the same recipe every month. It does not look at what you already own.

$$x_i = w_i \times C_t$$

x_i rupees that go into jar i this month

w_i target share of jar i (the shares add up to 100%)

C_t this month's instalment (₹10,000)

Example: Feb 2010: Nifty: $100.0\% \times ₹10,000 = ₹10,000.00$, Gold: $0.0\% \times ₹10,000 = ₹0.00$, Liquid: $0.0\% \times ₹10,000 = ₹0.00$.

4 Buy: put the money in the jars

Each piece of the instalment is added to its jar.

$$h_i \leftarrow h_i + x_i$$

h_i rupees in jar i (Nifty, Gold or Liquid)

$\leftarrow$ "becomes": the new value replaces the old one

x_i rupees bought for jar i this month

Example: Feb 2010: Nifty: $₹0.00 + ₹10,000.00 = ₹10,000.00$, Gold: $₹0.00 + ₹0.00 = ₹0.00$, Liquid: $₹0.00 + ₹0.00 = ₹0.00$.

5 Rebalance? Never

There is only one jar with money in it, so there is nothing to rebalance.

$$\text{rebalance} = \text{no} \quad (\text{every month})$$

rule `rebalance="none"` in `strategy.py`

Example: Feb 2010: split after buying = Nifty 100.0%, Gold 0.0%, Liquid 0.0%. Nothing is sold, so no trades.

6 Pay the trading fee

Every rupee bought or sold costs 0.1% (10 paise per ₹100), like a broker's charge. The fee is taken from all jars in proportion to their size, so the split does not change.

$$Fee = c \times \left(\sum_i buys_i + \sum_i sells_i \right)$$
$$h_i \leftarrow h_i \times \left(1 - \frac{Fee}{V_{start}} \right)$$

c cost rate = 0.1% = 0.001

$buys_i, sells_i$ rupees bought / sold in jar i this month

V_{start} total of all jars just before the fee

$1 - Fee/V_{start}$ the fraction of each jar that is kept

Example: Feb 2010: traded ₹10,000.00 (only the instalment was bought). $Fee = 0.001 \times ₹10,000.00 = ₹10.00$. Each jar keeps $1 - ₹10.00 \div ₹10,000.00 = 0.999000$ of its value → Nifty ₹9,990.00, Gold ₹0.00, Liquid ₹0.00.

7 The market moves

During the month each jar grows or shrinks by that asset's return. A +1% month turns ₹100 into ₹101; a -2% month turns it into ₹98.

$$h_i \leftarrow h_i \times (1 + r_{i,t})$$

$r_{i,t}$ the return of asset i in month t (from 00_raw_data/monthly_returns.csv)

Example: Feb 2010 returns: Nifty +0.93%, Gold +3.17%, Liquid +0.31%.
Nifty: ₹9,990.00 $\times (1 + 0.0093) = ₹10,083.18$. End-of-month total **₹10,083.18**.

8 Measure this month's return (TWR)

How well did the *strategy* do this month, ignoring the fact that you added new money? Compare the value at the end with the value right after the money went in.

$$TWR_t = \frac{V_{end}}{V_{start}} - 1$$

V_{end} total of all jars at the end of the month

V_{start} total just after buying (and any rebalancing), before the fee

TWR_t time-weighted return of month t (the fee counts as part of the loss)

Example: Feb 2010: ₹10,083.18 $\div ₹10,000.00 - 1 = +0.83\%$.

🔄 **Repeat steps 2-8 for every month**, Feb 2010 → Dec 2019 (119 times). Each month starts with last month's end-of-month jars. After the last month the jars hold **₹2,176,119**.

Money example: Mar 2010, all steps together

The second month, showing how the jar builds up: last month's ₹10,083.18 is still there, and the new ₹10,000 is added on top.

	Nifty	Gold	Liquid	Total
① Start of month (last month's end)	₹10,083.18	₹0.00	₹0.00	₹10,083.18
② Instalment ₹10,000 split into	₹10,000.00	₹0.00	₹0.00	₹10,000.00
③ Jars after buying	₹20,083.18	₹0.00	₹0.00	₹20,083.18
③ Share of total after buying	100.0%	0.0%	0.0%	100%
Target split w	100.0%	0.0%	0.0%	100%
④ Rebalance?	No			
⑤ Fee (0.1% of ₹10,000.00 traded)	taken from every jar in proportion			–₹10.00
⑤ Jars after fee	₹20,073.18	₹0.00	₹0.00	₹20,073.18
⑥ Market return this month r	+6.75%	–2.77%	+0.37%	
⑥ End of month	₹21,427.62	₹0.00	₹0.00	₹21,427.62

Nifty rose 6.75% in March 2010, and the *whole* ₹20,073.18 earned it, not only the new ₹10,000. The month's return: $₹21,427.62 \div ₹20,073.18 - 1 = +6.69\%$. Paid in so far: ₹20,000.

Scoring after 119 months

9a Score 1: yearly return on your money (XIRR)

Your instalments went in at different times, so a simple average is wrong. XIRR is the single yearly interest rate that, if a bank paid it on every instalment from the day you paid it, would give exactly your final value.

$$\sum_{k=0}^{118} C_k \times (1 + XIRR)^{(119-k)/12} = V_{final}$$

C_k instalment paid at the start of month k (₹10,000)

$(119 - k)/12$ years that instalment was invested until the end

V_{final} the value at the end of Dec 2019

Example: 119 instalments of ₹10,000 grew to **₹2,176,119**. The rate that makes both sides equal (found by trial and error, brentq) is **XIRR = 11.71%** a year. (For comparison, a 100% Liquid SIP gives 7.28%.)

9b Score 2: how bumpy is the ride (volatility)

Volatility measures how much the monthly return jumps around its average. Bigger jumps mean a scarier ride.

$$\sigma = \text{std}(TWR_1, \dots, TWR_{119}) \times \sqrt{12}$$

std standard deviation: the typical distance from the average

$\sqrt{12}$ turns a monthly figure into a yearly one

Example: monthly std = 4.42% $\rightarrow \sigma = 4.42\% \times 3.464 = 15.33\%$ a year.

9c Score 3: reward per unit of risk (Sharpe vs Liquid)

Liquid (a T-bill fund) earns interest with almost no risk. So the fair question is: how much *extra* return did the strategy earn above Liquid, per unit of bumpiness?

$$Sharpe_{vs\ Liquid} = \frac{12 \times \text{mean}(TWR_t - r_{Liquid,t})}{\sigma}$$

$r_{Liquid,t}$ Liquid's return in month t ($\approx$ the 91-day T-bill rate)

$12 \times \text{mean}(\dots)$ average extra return per year

σ volatility from score 2

Example: average extra return = 0.366% a month $\rightarrow \times 12 = 4.39\%$ a year; $4.39\% \div 15.33\% = 0.29$.
(Measured against 0% instead of Liquid it would be 0.76.)

9d Score 4: worst fall (maximum drawdown)

The biggest drop from a high point to a later low point, as if ₹1 had followed the strategy's monthly returns. It answers: "what is the worst loss I would have felt?"

$$G_t = \prod_{k \leq t} (1 + TWR_k) \quad MDD = \min_t \left(\frac{G_t}{\max_{k \leq t} G_k} - 1 \right)$$

G_t growth of ₹1 up to month t

$\max_{k \leq t} G_k$ the highest point reached so far

MDD the deepest fall below a previous high

Example: ₹1 grew to ₹1.2676 by Dec 2010, then fell to ₹0.9676 by Dec 2011: $0.9676 \div 1.2676 - 1 = -23.7\%$.

Paid in

₹1,190,000

Final value

₹2,176,119

XIRR

11.7%

Volatility

15.3%

Sharpe vs Liquid

0.29

Max drawdown

-23.7%

Strategy 2 Equal-weight SIP: $\frac{1}{3}$ each

The simplest way to diversify: split every ₹10,000 equally between Nifty, Gold and Liquid, and never sell anything.

Setting	This strategy
Where the target split w comes from	Fixed: $\frac{1}{3} / \frac{1}{3} / \frac{1}{3}$
How each instalment is split	Pro-rata ($\frac{1}{3}$ each)
When the whole portfolio is rebalanced	Never
Instalment, fee, market move, TWR, scoring	Same for all six strategies

The steps and their formulas

1 Choose the target split (fixed)

Each asset gets one third, a split that needs no forecasts at all.

$$w = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

w_i target share of jar i ; the three shares add up to 1 (100%)

Example: the same target is used in every one of the 119 months. Nothing is learned from data.

2 The instalment arrives

On the 1st of every month you pay in the same amount. The formula allows the amount to grow every year (a “step-up”), but in this project the growth rate is 0, so it is always ₹10,000.

$$C_t = A \times (1 + g)^{\lfloor t/12 \rfloor}$$

C_t the instalment paid in month t

t the month number, counting from 0 (Feb 2010 is $t = 0$, Dec 2019 is $t = 118$)

A the starting instalment = ₹10,000

g yearly step-up of the instalment = 0 (no step-up)

$\lfloor t/12 \rfloor$ number of full years passed ($t \div 12$, rounded down)

Example: Feb 2010: $C = 10,000 \times (1 + 0)^0 = \text{₹10,000}$. Because $g = 0$, every one of the 119 months is ₹10,000, so the total paid in is $119 \times \text{₹10,000} = \text{₹1,190,000}$.

3 Split the instalment (pro-rata)

The ₹10,000 is cut into pieces in exactly the target proportions, like cutting a cake using the same recipe every month. It does not look at what you already own.

$$x_i = w_i \times C_t$$

x_i rupees that go into jar i this month

w_i target share of jar i (the shares add up to 100%)

C_t this month's instalment (₹10,000)

Example: Feb 2010: Nifty: $33.3\% \times ₹10,000 = ₹3,333.33$, Gold: $33.3\% \times ₹10,000 = ₹3,333.33$, Liquid: $33.3\% \times ₹10,000 = ₹3,333.33$.

4 Buy: put the money in the jars

Each piece of the instalment is added to its jar.

$$h_i \leftarrow h_i + x_i$$

h_i rupees in jar i (Nifty, Gold or Liquid)

$\leftarrow$ "becomes": the new value replaces the old one

x_i rupees bought for jar i this month

Example: Feb 2010: Nifty: $₹0.00 + ₹3,333.33 = ₹3,333.33$, Gold: $₹0.00 + ₹3,333.33 = ₹3,333.33$, Liquid: $₹0.00 + ₹3,333.33 = ₹3,333.33$.

5 Rebalance? Never

The jars are allowed to drift. If Nifty booms, its share grows above $\frac{1}{3}$ and stays there. The only thing that pulls it back is that each new instalment is split equally.

$$\text{rebalance} = \text{no} \quad (\text{every month})$$

rule rebalance="none" in strategy.py

Example: Mar 2010: split after buying = Nifty 33.2%, Gold 33.6%, Liquid 33.1%. Nothing is sold, so no trades.

6 Pay the trading fee

Every rupee bought or sold costs 0.1% (10 paise per ₹100), like a broker's charge. The fee is taken from all jars in proportion to their size, so the split does not change.

$$Fee = c \times \left(\sum_i buys_i + \sum_i sells_i \right)$$

$$h_i \leftarrow h_i \times \left(1 - \frac{Fee}{V_{start}} \right)$$

c cost rate = 0.1% = 0.001

$buys_i, sells_i$ rupees bought / sold in jar i this month

V_{start} total of all jars just before the fee

$1 - Fee/V_{start}$ the fraction of each jar that is kept

Example: Feb 2010: traded ₹10,000.00 (only the instalment was bought). $Fee = 0.001 \times ₹10,000.00 = ₹10.00$. Each jar keeps $1 - ₹10.00 \div ₹10,000.00 = 0.999000$ of its value → Nifty ₹3,330.00, Gold ₹3,330.00, Liquid ₹3,330.00.

7 The market moves

During the month each jar grows or shrinks by that asset's return. A +1% month turns ₹100 into ₹101; a -2% month turns it into ₹98.

$$h_i \leftarrow h_i \times (1 + r_{i,t})$$

$r_{i,t}$ the return of asset i in month t (from 00_raw_data/monthly_returns.csv)

Example: Feb 2010 returns: Nifty +0.93%, Gold +3.17%, Liquid +0.31%.

Nifty: ₹3,330.00 × (1 + 0.0093) = ₹3,361.06, Gold: ₹3,330.00 × (1 + 0.0317) = ₹3,435.61, Liquid: ₹3,330.00 × (1 + 0.0031) = ₹3,340.49. End-of-month total **₹10,137.16**.

8 Measure this month's return (TWR)

How well did the *strategy* do this month, ignoring the fact that you added new money? Compare the value at the end with the value right after the money went in.

$$TWR_t = \frac{V_{end}}{V_{start}} - 1$$

V_{end} total of all jars at the end of the month

V_{start} total just after buying (and any rebalancing), before the fee

TWR_t time-weighted return of month t (the fee counts as part of the loss)

Example: Feb 2010: ₹10,137.16 ÷ ₹10,000.00 - 1 = **+1.37%**.

🔄 **Repeat steps 2-8 for every month**, Feb 2010 → Dec 2019 (119 times). Each month starts with last month's end-of-month jars. After the last month the jars hold **₹1,850,460**.

Money example: Mar 2010, all steps together

The second month. After February, Gold's jar is the biggest because gold rose most; the new ₹10,000 is still split equally.

	Nifty	Gold	Liquid	Total
① Start of month (last month's end)	₹3,361.06	₹3,435.61	₹3,340.49	₹10,137.16
② Instalment ₹10,000 split into	₹3,333.33	₹3,333.33	₹3,333.33	₹10,000.00
③ Jars after buying	₹6,694.39	₹6,768.94	₹6,673.82	₹20,137.16
③ Share of total after buying	33.2%	33.6%	33.1%	100%
Target split w	33.3%	33.3%	33.3%	100%
④ Rebalance?	No			
⑤ Fee (0.1% of ₹10,000.00 traded)	taken from every jar in proportion			–₹10.00
⑤ Jars after fee	₹6,691.07	₹6,765.58	₹6,670.51	₹20,127.16
⑥ Market return this month r	+6.75%	–2.77%	+0.37%	
⑥ End of month	₹7,142.55	₹6,578.00	₹6,695.06	₹20,415.61

In March Nifty rose 6.75% but gold fell 2.77%, so the jars moved in opposite directions. That is diversification at work. The month's return: $₹20,415.61 \div ₹20,137.16 - 1 = +1.38\%$. Paid in so far: ₹20,000.

Scoring after 119 months

9a Score 1: yearly return on your money (XIRR)

Your instalments went in at different times, so a simple average is wrong. XIRR is the single yearly interest rate that, if a bank paid it on every instalment from the day you paid it, would give exactly your final value.

$$\sum_{k=0}^{118} C_k \times (1 + XIRR)^{(119-k)/12} = V_{final}$$

C_k instalment paid at the start of month k (₹10,000)

$(119 - k)/12$ years that instalment was invested until the end

V_{final} the value at the end of Dec 2019

Example: 119 instalments of ₹10,000 grew to **₹1,850,460**. The rate that makes both sides equal (found by trial and error, brentq) is **XIRR = 8.62%** a year. (For comparison, a 100% Liquid SIP gives 7.28%.)

9b Score 2: how bumpy is the ride (volatility)

Volatility measures how much the monthly return jumps around its average. Bigger jumps mean a scarier ride.

$$\sigma = \text{std}(TWR_1, \dots, TWR_{119}) \times \sqrt{12}$$

std standard deviation: the typical distance from the average

$\sqrt{12}$ turns a monthly figure into a yearly one

Example: monthly std = 1.87% $\rightarrow \sigma = 1.87\% \times 3.464 = 6.49\%$ a year.

9c Score 3: reward per unit of risk (Sharpe vs Liquid)

Liquid (a T-bill fund) earns interest with almost no risk. So the fair question is: how much *extra* return did the strategy earn above Liquid, per unit of bumpiness?

$$Sharpe_{vs\ Liquid} = \frac{12 \times \text{mean}(TWR_t - r_{Liquid,t})}{\sigma}$$

$r_{Liquid,t}$ Liquid's return in month t ($\approx$ the 91-day T-bill rate)

$12 \times \text{mean}(\dots)$ average extra return per year

σ volatility from score 2

Example: average extra return = 0.151% a month $\rightarrow \times 12 = 1.81\%$ a year; $1.81\% \div 6.49\% = 0.28$.
(Measured against 0% instead of Liquid it would be 1.40.)

9d Score 4: worst fall (maximum drawdown)

The biggest drop from a high point to a later low point, as if ₹1 had followed the strategy's monthly returns. It answers: "what is the worst loss I would have felt?"

$$G_t = \prod_{k \leq t} (1 + TWR_k) \quad MDD = \min_t \left(\frac{G_t}{\max_{k \leq t} G_k} - 1 \right)$$

G_t growth of ₹1 up to month t

$\max_{k \leq t} G_k$ the highest point reached so far

MDD the deepest fall below a previous high

Example: ₹1 grew to ₹1.4384 by Nov 2012, then fell to ₹1.3445 by Jun 2013: $1.3445 \div 1.4384 - 1 = -6.5\%$.

Paid in

₹1,190,000

Final value

₹1,850,460

XIRR

8.6%

Volatility

6.5%

Sharpe vs Liquid

0.28

Max drawdown

-6.5%

Strategy 3 Fixed 60/20/20 SIP, rebalanced every year

A classic balanced mix: 60% Nifty for growth, 20% Gold as a hedge and 20% Liquid as a cushion. Once a year the jars are reset to 60/20/20.

Setting	This strategy
Where the target split w comes from	Fixed: 60 / 20 / 20
How each instalment is split	Pro-rata (60/20/20)
When the whole portfolio is rebalanced	Every 12 months (calendar)
Instalment, fee, market move, TWR, scoring	Same for all six strategies

The steps and their formulas

1 Choose the target split (fixed)

A conventional balanced split chosen by hand, not fitted to the data.

$$w = (0.60, 0.20, 0.20)$$

w_i target share of jar i ; the three shares add up to 1 (100%)

Example: the same target is used in every one of the 119 months. Nothing is learned from data.

2 The instalment arrives

On the 1st of every month you pay in the same amount. The formula allows the amount to grow every year (a “step-up”), but in this project the growth rate is 0, so it is always ₹10,000.

$$C_t = A \times (1 + g)^{\lfloor t/12 \rfloor}$$

C_t the instalment paid in month t

t the month number, counting from 0 (Feb 2010 is $t = 0$, Dec 2019 is $t = 118$)

A the starting instalment = ₹10,000

g yearly step-up of the instalment = 0 (no step-up)

$\lfloor t/12 \rfloor$ number of full years passed ($t \div 12$, rounded down)

Example: Feb 2010: $C = 10,000 \times (1 + 0)^0 = \text{₹}10,000$. Because $g = 0$, every one of the 119 months is ₹10,000, so the total paid in is $119 \times \text{₹}10,000 = \text{₹}1,190,000$.

3 Split the instalment (pro-rata)

The ₹10,000 is cut into pieces in exactly the target proportions, like cutting a cake using the same recipe every month. It does not look at what you already own.

$$x_i = w_i \times C_t$$

x_i rupees that go into jar i this month

w_i target share of jar i (the shares add up to 100%)

C_t this month's instalment (₹10,000)

Example: Feb 2010: Nifty: $60.0\% \times ₹10,000 = ₹6,000.00$, Gold: $20.0\% \times ₹10,000 = ₹2,000.00$, Liquid: $20.0\% \times ₹10,000 = ₹2,000.00$.

4 Buy: put the money in the jars

Each piece of the instalment is added to its jar.

$$h_i \leftarrow h_i + x_i$$

h_i rupees in jar i (Nifty, Gold or Liquid)

$\leftarrow$ "becomes": the new value replaces the old one

x_i rupees bought for jar i this month

Example: Feb 2010: Nifty: $₹0.00 + ₹6,000.00 = ₹6,000.00$, Gold: $₹0.00 + ₹2,000.00 = ₹2,000.00$, Liquid: $₹0.00 + ₹2,000.00 = ₹2,000.00$.

5 Rebalance? Once a year (calendar rule)

Every 12 months the jars are put back to exactly 60/20/20: sell some of the jars that grew too big and buy the jars that fell behind. The total does not change, money only moves between jars. In the other 11 months nothing happens here.

$$\text{rebalance if } n_{\text{since}} \geq 12$$

$$\text{trade}_i = w_i \times \sum_j h_j - h_i \quad h_i \leftarrow w_i \times \sum_j h_j$$

n_{since} months since the last rebalance (counts 1, 2, ... 12)

$\sum_j h_j$ total of all jars

trade_i rupees to buy (+) or sell (-) in jar i

Example: Feb 2010 is month 1, so the first rebalance is month 12 = Jan 2011. After buying, the total is ₹133,625.34:
 Nifty: $60\% \times ₹133,625.34 - ₹81,757.59 = -₹1,582.39$ (sell)
 Gold: $20\% \times ₹133,625.34 - ₹27,208.57 = -₹483.50$ (sell)
 Liquid: $20\% \times ₹133,625.34 - ₹24,659.18 = ₹2,065.88$ (buy).
 Sold ₹2,065.88, bought the same amount. It happened 9 times in 10 years (every January 2011–2019).

6 Pay the trading fee

Every rupee bought or sold costs 0.1% (10 paise per ₹100), like a broker's charge. The fee is taken from all jars in proportion to their size, so the split does not change.

$$Fee = c \times \left(\sum_i buys_i + \sum_i sells_i \right)$$
$$h_i \leftarrow h_i \times \left(1 - \frac{Fee}{V_{start}} \right)$$

c cost rate = 0.1% = 0.001

$buys_i, sells_i$ rupees bought / sold in jar i this month

V_{start} total of all jars just before the fee

$1 - Fee/V_{start}$ the fraction of each jar that is kept

Example: Feb 2010: traded ₹10,000.00 (only the instalment was bought). $Fee = 0.001 \times ₹10,000.00 = ₹10.00$. Each jar keeps $1 - ₹10.00 \div ₹10,000.00 = 0.999000$ of its value → Nifty ₹5,994.00, Gold ₹1,998.00, Liquid ₹1,998.00.

7 The market moves

During the month each jar grows or shrinks by that asset's return. A +1% month turns ₹100 into ₹101; a -2% month turns it into ₹98.

$$h_i \leftarrow h_i \times (1 + r_{i,t})$$

$r_{i,t}$ the return of asset i in month t (from 00_raw_data/monthly_returns.csv)

Example: Feb 2010 returns: Nifty +0.93%, Gold +3.17%, Liquid +0.31%.

Nifty: ₹5,994.00 $\times (1 + 0.0093) = ₹6,049.91$, Gold: ₹1,998.00 $\times (1 + 0.0317) = ₹2,061.36$, Liquid: ₹1,998.00 $\times (1 + 0.0031) = ₹2,004.29$. End-of-month total **₹10,115.57**.

8 Measure this month's return (TWR)

How well did the *strategy* do this month, ignoring the fact that you added new money? Compare the value at the end with the value right after the money went in.

$$TWR_t = \frac{V_{end}}{V_{start}} - 1$$

V_{end} total of all jars at the end of the month

V_{start} total just after buying (and any rebalancing), before the fee

TWR_t time-weighted return of month t (the fee counts as part of the loss)

Example: Feb 2010: ₹10,115.57 $\div$ ₹10,000.00 $- 1 = +1.16\%$.

🔄 **Repeat steps 2-8 for every month**, Feb 2010 → Dec 2019 (119 times). Each month starts with last month's end-of-month jars. After the last month the jars hold **₹1,990,520**.

Money example: Jan 2011, all steps together

The 12th month, the first yearly rebalance. Nifty had grown to 61.2% and Liquid had fallen to 18.5%, so money is moved from Nifty and Gold into Liquid.

	Nifty	Gold	Liquid	Total
① Start of month (last month's end)	₹75,757.59	₹25,208.57	₹22,659.18	₹123,625.34
② Instalment ₹10,000 split into	₹6,000.00	₹2,000.00	₹2,000.00	₹10,000.00
③ Jars after buying	₹81,757.59	₹27,208.57	₹24,659.18	₹133,625.34
③ Share of total after buying	61.2%	20.4%	18.5%	100%
Target split w	60.0%	20.0%	20.0%	100%
④ Rebalance trades (– = sell)	–₹1,582.39	–₹483.50	₹2,065.88	
④ Jars after rebalancing	₹80,175.20	₹26,725.07	₹26,725.07	₹133,625.34
⑤ Fee (0.1% of ₹14,131.77 traded)	taken from every jar in proportion			–₹14.13
⑤ Jars after fee	₹80,166.72	₹26,722.24	₹26,722.24	₹133,611.21
⑥ Market return this month r	–10.14%	–3.77%	+0.61%	
⑥ End of month	₹72,038.91	₹25,715.74	₹26,885.35	₹124,640.01

Nifty then fell 10.14% in January 2011. Because ₹1,582 had just been moved out of Nifty into Liquid, the fall hurt a little less. The fee was ₹14.13 instead of ₹10, because the rebalance trades also pay 0.1%. The month's return: $₹124,640.01 \div ₹133,625.34 - 1 = -6.72\%$. Paid in so far: ₹120,000.

Scoring after 119 months

9a Score 1: yearly return on your money (XIRR)

Your instalments went in at different times, so a simple average is wrong. XIRR is the single yearly interest rate that, if a bank paid it on every instalment from the day you paid it, would give exactly your final value.

$$\sum_{k=0}^{118} C_k \times (1 + XIRR)^{(119-k)/12} = V_{final}$$

C_k instalment paid at the start of month k (₹10,000)

$(119 - k)/12$ years that instalment was invested until the end

V_{final} the value at the end of Dec 2019

Example: 119 instalments of ₹10,000 grew to **₹1,990,520**. The rate that makes both sides equal (found by trial and error, brentq) is **XIRR = 10.02%** a year. (For comparison, a 100% Liquid SIP gives 7.28%.)

9b Score 2: how bumpy is the ride (volatility)

Volatility measures how much the monthly return jumps around its average. Bigger jumps mean a scarier ride.

$$\sigma = \text{std}(TWR_1, \dots, TWR_{119}) \times \sqrt{12}$$

std standard deviation: the typical distance from the average

$\sqrt{12}$ turns a monthly figure into a yearly one

Example: monthly std = 2.54% $\rightarrow \sigma = 2.54\% \times 3.464 = 8.81\%$ a year.

9c Score 3: reward per unit of risk (Sharpe vs Liquid)

Liquid (a T-bill fund) earns interest with almost no risk. So the fair question is: how much *extra* return did the strategy earn above Liquid, per unit of bumpiness?

$$Sharpe_{vs\ Liquid} = \frac{12 \times \text{mean}(TWR_t - r_{Liquid,t})}{\sigma}$$

$r_{Liquid,t}$ Liquid's return in month t ($\approx$ the 91-day T-bill rate)

$12 \times \text{mean}(\dots)$ average extra return per year

σ volatility from score 2

Example: average extra return = 0.243% a month $\rightarrow \times 12 = 2.91\%$ a year; $2.91\% \div 8.81\% = 0.33$.
(Measured against 0% instead of Liquid it would be 1.15.)

9d Score 4: worst fall (maximum drawdown)

The biggest drop from a high point to a later low point, as if ₹1 had followed the strategy's monthly returns. It answers: "what is the worst loss I would have felt?"

$$G_t = \prod_{k \leq t} (1 + TWR_k) \quad MDD = \min_t \left(\frac{G_t}{\max_{k \leq t} G_k} - 1 \right)$$

G_t growth of ₹1 up to month t

$\max_{k \leq t} G_k$ the highest point reached so far

MDD the deepest fall below a previous high

Example: ₹1 grew to ₹1.8224 by Jan 2015, then fell to ₹1.6594 by Feb 2016: $1.6594 \div 1.8224 - 1 = -8.9\%$.

Paid in

₹1,190,000

Final value

₹1,990,520

XIRR

10.0%

Volatility

8.8%

Sharpe vs Liquid

0.33

Max drawdown

-8.9%

Strategy 4 Risk-parity SIP

An optimiser chooses the split so that each asset contributes the same amount of risk. It re-learns the split every year from the previous 10 years.

Setting	This strategy
Where the target split w comes from	Optimiser: risk parity (re-fit yearly)
How each instalment is split	Smart (fill the gaps first)
When the whole portfolio is rebalanced	Only if drift > 5% (band)
Instalment, fee, market move, TWR, scoring	Same for all six strategies

The steps and their formulas

1a Look back 10 years: average return μ and covariance Σ

Before each yearly decision the optimiser studies only the **past** 120 months. It never sees the future. For the first SIP month it uses Feb 2000 – Jan 2010.

$$\mu_i = \frac{1}{120} \sum_{k=1}^{120} r_{i,t-k} \quad \Sigma_{ij} = \frac{1}{119} \sum_{k=1}^{120} (r_{i,t-k} - \mu_i)(r_{j,t-k} - \mu_j)$$

μ_i average monthly return of asset i over the last 120 months

Σ_{ij} covariance: how assets i and j move together (Σ_{ii} is asset i 's variance = volatility²)

$r_{i,t-k}$ return of asset i , k months ago

Example: Feb 2000 – Jan 2010 (as yearly figures): average return Nifty 16.8%, Gold 15.3%, Liquid 6.2%; volatility Nifty 28.0%, Gold 16.8%, Liquid 0.5%. Liquid barely moves (volatility 0.5%): it looks almost risk-free to the optimiser.

1b Risk parity: every asset adds the same amount of risk

The idea: don't let one asset dominate the risk. Each asset's *risk contribution* (its weight $\times$ how much it moves the whole portfolio) should be equal. The optimiser picks the weights that make the three contributions as equal as possible.

$$RC_i = w_i \times (\Sigma w)_i \quad \min_w \sum_i (RC_i - \overline{RC})^2$$

$(\Sigma w)_i$ how much asset i moves together with the whole portfolio

RC_i risk contribution of asset i (they add up to the portfolio variance)

$\overline{RC}$ the average of the three contributions

Example: Feb 2010 result: **Nifty 10.7% / Gold 19.3% / Liquid 70.0%**. Shares of risk: Nifty 46%, Gold 55%, Liquid –1%. Liquid can't reach an equal share because it hits the 70% cap (its share is slightly negative because it tends to rise when Nifty falls).

1c Limits every optimiser must obey

To stop the optimiser putting everything in one asset, each weight is kept between 10% and 70%, and the weights must add up to 100%. The best weights are found by a standard solver (SLSQP) that tries weights until the goal cannot be improved.

$$0.10 \leq w_i \leq 0.70 \quad \sum_i w_i = 1$$

0.10, 0.70 lowest / highest allowed share of any asset

Example: Liquid ends up at the 70% cap in every year. Without the cap, risk parity would put 96.2% in Liquid.

1d Re-learn every year (walk-forward)

Every 12 months (each February) the 10-year window moves forward one year and steps 1a-1c are repeated. The new target is used for the next 12 months.

$$w^{(Feb\ Y)} = \text{optimiser}(r_{Feb\ Y-10}, \dots, r_{Jan\ Y})$$

Y the year of the decision (2010, 2011, ..., 2019)

Example: Feb 2011 uses Feb 2001 – Jan 2011, ..., Feb 2019 uses Feb 2009 – Jan 2019. 10 decisions in total.

2 The instalment arrives

On the 1st of every month you pay in the same amount. The formula allows the amount to grow every year (a “step-up”), but in this project the growth rate is 0, so it is always ₹10,000.

$$C_t = A \times (1 + g)^{\lfloor t/12 \rfloor}$$

C_t the instalment paid in month t

t the month number, counting from 0 (Feb 2010 is $t = 0$, Dec 2019 is $t = 118$)

A the starting instalment = ₹10,000

g yearly step-up of the instalment = 0 (no step-up)

$\lfloor t/12 \rfloor$ number of full years passed ($t \div 12$, rounded down)

Example: Feb 2010: $C = 10,000 \times (1 + 0)^0 = \text{₹10,000}$. Because $g = 0$, every one of the 119 months is ₹10,000, so the total paid in is $119 \times \text{₹10,000} = \text{₹1,190,000}$.

3 Split the instalment (smart)

Instead of copying the recipe, the ₹10,000 is sent *first* to the jars that are below their target. Think of topping up the emptiest glasses first. This pulls the portfolio back towards the target **without selling anything**.

$$V = \sum_i h_i + C_t \quad \text{gap}_i = \max(w_i V - h_i, 0) \quad G = \sum_i \text{gap}_i$$

$$\text{If } G \geq C_t : \quad x_i = \text{gap}_i \times \frac{C_t}{G}$$

$$\text{If } G < C_t : \quad x_i = \text{gap}_i + w_i \times (C_t - G)$$

h_i rupees already in jar i before buying

V total wealth after this month's instalment

$w_i V$ what jar i *should* hold

gap_i how many rupees jar i is short of its target (0 if it is already above)

G all the gaps added together

x_i rupees that go into jar i this month

Example: Feb 2010: jars hold Nifty ₹0.00, Gold ₹0.00, Liquid ₹0.00; $V = ₹0.00 + ₹10,000 = ₹10,000.00$.

Nifty: $\max(10.66\% \times ₹10,000.00 - ₹0.00, 0) = \max(₹1,066.12 - ₹0.00, 0) = ₹1,066.12$

Gold: $\max(19.34\% \times ₹10,000.00 - ₹0.00, 0) = \max(₹1,933.88 - ₹0.00, 0) = ₹1,933.88$

Liquid: $\max(70.00\% \times ₹10,000.00 - ₹0.00, 0) = \max(₹7,000.00 - ₹0.00, 0) = ₹7,000.00$

$G = ₹10,000.00$: all three jars were below target, so the gaps add up to exactly the ₹10,000 available and each jar gets exactly its gap $\rightarrow$ Nifty ₹1,066.12, Gold ₹1,933.88, Liquid ₹7,000.00. (In the very first month all jars are empty, so the gaps are exactly $w \times ₹10,000$.)

4 Buy: put the money in the jars

Each piece of the instalment is added to its jar.

$$h_i \leftarrow h_i + x_i$$

h_i rupees in jar i (Nifty, Gold or Liquid)

$\leftarrow$ "becomes": the new value replaces the old one

x_i rupees bought for jar i this month

Example: Feb 2010: Nifty: ₹0.00 + ₹1,066.12 = ₹1,066.12, Gold: ₹0.00 + ₹1,933.88 = ₹1,933.88, Liquid: ₹0.00 + ₹7,000.00 = ₹7,000.00.

5 Rebalance? Only if something drifts more than 5% (band rule)

After buying, the engine measures how far each jar's share is from its target. If the biggest gap (the *drift*) is more than 5 percentage points, the jars are reset to the target exactly as in the calendar rule. Otherwise nothing is sold.

$$drift = \max_i \left| \frac{h_i}{\sum_j h_j} - w_i \right| \quad \text{rebalance if } drift > 0.05$$

$$trade_i = w_i \times \sum_j h_j - h_i$$

$h_i / \sum_j h_j$ jar i's actual share of the total

$|\dots|$ size of the gap, ignoring its sign

$\max_i$ the biggest of the three gaps

0.05 the 5% band

Example: Feb 2011: Nifty $|11.02\% - 11.02\%| = 0.00\%$, Gold $|18.98\% - 18.98\%| = 0.00\%$, Liquid $|70.00\% - 70.00\%| = 0.00\%$. Drift = 0.00%, below 5% → **no rebalance**. The smart split already filled the gaps. Over 10 years the band was never crossed (0 rebalances), so this strategy never sold anything.

6 Pay the trading fee

Every rupee bought or sold costs 0.1% (10 paise per ₹100), like a broker's charge. The fee is taken from all jars in proportion to their size, so the split does not change.

$$Fee = c \times \left(\sum_i buys_i + \sum_i sells_i \right)$$

$$h_i \leftarrow h_i \times \left(1 - \frac{Fee}{V_{start}} \right)$$

c cost rate = 0.1% = 0.001

$buys_i, sells_i$ rupees bought / sold in jar i this month

V_{start} total of all jars just before the fee

$1 - Fee/V_{start}$ the fraction of each jar that is kept

Example: Feb 2010: traded ₹10,000.00 (only the instalment was bought). $Fee = 0.001 \times ₹10,000.00 = ₹10.00$. Each jar keeps $1 - ₹10.00 \div ₹10,000.00 = 0.999000$ of its value → Nifty ₹1,065.05, Gold ₹1,931.95, Liquid ₹6,993.00.

7 The market moves

During the month each jar grows or shrinks by that asset's return. A +1% month turns ₹100 into ₹101; a −2% month turns it into ₹98.

$$h_i \leftarrow h_i \times (1 + r_{i,t})$$

$r_{i,t}$ the return of asset i in month t (from 00_raw_data/monthly_returns.csv)

Example: Feb 2010 returns: Nifty +0.93%, Gold +3.17%, Liquid +0.31%.

Nifty: ₹1,065.05 × (1 + 0.0093) = ₹1,074.99, Gold: ₹1,931.95 × (1 + 0.0317) = ₹1,993.22, Liquid: ₹6,993.00 × (1 + 0.0031) = ₹7,015.03. End-of-month total **₹10,083.23**.

8 Measure this month's return (TWR)

How well did the *strategy* do this month, ignoring the fact that you added new money? Compare the value at the end with the value right after the money went in.

$$TWR_t = \frac{V_{end}}{V_{start}} - 1$$

V_{end} total of all jars at the end of the month

V_{start} total just after buying (and any rebalancing), before the fee

TWR_t time-weighted return of month t (the fee counts as part of the loss)

Example: Feb 2010: ₹10,083.23 ÷ ₹10,000.00 − 1 = **+0.83%**.

🔄 **Repeat steps 2-8 for every month**, Feb 2010 → Dec 2019 (119 times). Each month starts with last month's end-of-month jars. After the last month the jars hold **₹1,778,154**.

The smart split in a later month (the jars are no longer empty)

3 Split the instalment (smart)

Instead of copying the recipe, the ₹10,000 is sent *first* to the jars that are below their target. Think of topping up the emptiest glasses first. This pulls the portfolio back towards the target **without selling anything**.

$$V = \sum_i h_i + C_t \quad \text{gap}_i = \max(w_i V - h_i, 0) \quad G = \sum_i \text{gap}_i$$

$$\begin{aligned} \text{If } G \geq C_t : \quad x_i &= \text{gap}_i \times \frac{C_t}{G} \\ \text{If } G < C_t : \quad x_i &= \text{gap}_i + w_i \times (C_t - G) \end{aligned}$$

h_i rupees already in jar i before buying

V total wealth after this month's instalment

$w_i V$ what jar i *should* hold

gap_i how many rupees jar i is short of its target (0 if it is already above)

G all the gaps added together

x_i rupees that go into jar i this month

Example: Feb 2011: jars hold Nifty ₹12,177.99, Gold ₹23,656.70, Liquid ₹89,523.84; $V = ₹125,358.53 + ₹10,000 = ₹135,358.53$.

Nifty: $\max(11.02\% \times ₹135,358.53 - ₹12,177.99, 0) = \max(₹14,910.56 - ₹12,177.99, 0) = ₹2,732.57$

Gold: $\max(18.98\% \times ₹135,358.53 - ₹23,656.70, 0) = \max(₹25,697.00 - ₹23,656.70, 0) = ₹2,040.30$

Liquid: $\max(70.00\% \times ₹135,358.53 - ₹89,523.84, 0) = \max(₹94,750.97 - ₹89,523.84, 0) =$

₹5,227.13

$G = ₹10,000.00$: all three jars were below target, so the gaps add up to exactly the ₹10,000 available and each jar gets exactly its gap → Nifty ₹2,732.57, Gold ₹2,040.30, Liquid ₹5,227.13.

Money example: Feb 2011, all steps together

February 2011: the optimiser has just re-learned the target from Feb 2001 – Jan 2011 and the smart split fills the gaps.

	Nifty	Gold	Liquid	Total
① Start of month (last month's end)	₹12,177.99	₹23,656.70	₹89,523.84	₹125,358.53
② Instalment ₹10,000 split into	₹2,732.57	₹2,040.30	₹5,227.13	₹10,000.00
③ Jars after buying	₹14,910.56	₹25,697.00	₹94,750.97	₹135,358.53
③ Share of total after buying	11.0%	19.0%	70.0%	100%
Target split w	11.0%	19.0%	70.0%	100%
④ Rebalance?	No			
⑤ Fee (0.1% of ₹10,000.00 traded)	taken from every jar in proportion			–₹10.00
⑤ Jars after fee	₹14,909.46	₹25,695.10	₹94,743.97	₹135,348.53
⑥ Market return this month r	–3.03%	+3.95%	+0.55%	
⑥ End of month	₹14,458.09	₹26,709.42	₹95,265.79	₹136,433.29

The new money went mostly where it was needed, so the split after buying equals the target and no selling was necessary. In the market move Nifty fell 3.03% while Gold rose 3.95%. The month's return: ₹136,433.29 ÷ ₹135,358.53 – 1 = **+0.79%**. Paid in so far: ₹130,000.

Scoring after 119 months

9a Score 1: yearly return on your money (XIRR)

Your instalments went in at different times, so a simple average is wrong. XIRR is the single yearly interest rate that, if a bank paid it on every instalment from the day you paid it, would give exactly your final value.

$$\sum_{k=0}^{118} C_k \times (1 + XIRR)^{(119-k)/12} = V_{final}$$

C_k instalment paid at the start of month k (₹10,000)

$(119 - k)/12$ years that instalment was invested until the end

V_{final} the value at the end of Dec 2019

Example: 119 instalments of ₹10,000 grew to **₹1,778,154**. The rate that makes both sides equal (found by trial and error, brentq) is **XIRR = 7.86%** a year. (For comparison, a 100% Liquid SIP gives 7.28%.)

9b Score 2: how bumpy is the ride (volatility)

Volatility measures how much the monthly return jumps around its average. Bigger jumps mean a scarier ride.

$$\sigma = \text{std}(TWR_1, \dots, TWR_{119}) \times \sqrt{12}$$

std standard deviation: the typical distance from the average

$\sqrt{12}$ turns a monthly figure into a yearly one

Example: monthly std = 0.90% → $\sigma = 0.90\% \times 3.464 = 3.12\%$ a year.

9c Score 3: reward per unit of risk (Sharpe vs Liquid)

Liquid (a T-bill fund) earns interest with almost no risk. So the fair question is: how much *extra* return did the strategy earn above Liquid, per unit of bumpiness?

$$Sharpe_{vs\ Liquid} = \frac{12 \times \text{mean}(TWR_t - r_{Liquid,t})}{\sigma}$$

$r_{Liquid,t}$ Liquid's return in month t ($\approx$ the 91-day T-bill rate)

$12 \times \text{mean}(\dots)$ average extra return per year

σ volatility from score 2

Example: average extra return = 0.069% a month → $\times 12 = 0.82\%$ a year; $0.82\% \div 3.12\% = 0.26$.
(Measured against 0% instead of Liquid it would be 2.59.)

9d Score 4: worst fall (maximum drawdown)

The biggest drop from a high point to a later low point, as if ₹1 had followed the strategy's monthly returns. It answers: "what is the worst loss I would have felt?"

$$G_t = \prod_{k \leq t} (1 + TWR_k) \quad MDD = \min_t \left(\frac{G_t}{\max_{k \leq t} G_k} - 1 \right)$$

G_t growth of ₹1 up to month t

$\max_{k \leq t} G_k$ the highest point reached so far

MDD the deepest fall below a previous high

Example: ₹1 grew to ₹1.2298 by Nov 2011, then fell to ₹1.2089 by Dec 2011: $1.2089 \div 1.2298 - 1 = -1.7\%$.

Paid in

₹1,190,000

Final value

₹1,778,154

XIRR

7.9%

Volatility

3.1%

Sharpe vs Liquid

0.26

Max drawdown

-1.7%

Strategy 5 Max-Sharpe SIP

An optimiser chooses the split with the highest expected return per unit of risk, re-learned every year from the previous 10 years.

Setting	This strategy
Where the target split w comes from	Optimiser: max Sharpe (re-fit yearly)
How each instalment is split	Smart (fill the gaps first)
When the whole portfolio is rebalanced	Only if drift > 5% (band)
Instalment, fee, market move, TWR, scoring	Same for all six strategies

The steps and their formulas

1a Look back 10 years: average return μ and covariance Σ

Before each yearly decision the optimiser studies only the **past** 120 months. It never sees the future. For the first SIP month it uses Feb 2000 – Jan 2010.

$$\mu_i = \frac{1}{120} \sum_{k=1}^{120} r_{i,t-k} \quad \Sigma_{ij} = \frac{1}{119} \sum_{k=1}^{120} (r_{i,t-k} - \mu_i)(r_{j,t-k} - \mu_j)$$

μ_i average monthly return of asset i over the last 120 months

Σ_{ij} covariance: how assets i and j move together (Σ_{ii} is asset i 's variance = volatility²)

$r_{i,t-k}$ return of asset i , k months ago

Example: Feb 2000 – Jan 2010 (as yearly figures): average return Nifty 16.8%, Gold 15.3%, Liquid 6.2%; volatility Nifty 28.0%, Gold 16.8%, Liquid 0.5%. Liquid barely moves (volatility 0.5%): it looks almost risk-free to the optimiser.

1b Max Sharpe: the most return per unit of risk

The optimiser tries every allowed mix and keeps the one with the highest expected return divided by expected risk.

$$\max_w \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}}$$

$w^\top \mu$ expected monthly return of the mix = $\sum w_i \mu_i$

$\sqrt{w^\top \Sigma w}$ expected monthly volatility of the mix

$\max_w$ choose the weights w that make this ratio largest

Example: Feb 2010 result: **Nifty 10.0% / Gold 20.0% / Liquid 70.0%**, ratio (per year) = $9.1\% \div 4.5\% = 2.03$. Compare 60/20/20: $14.4\% \div 17.4\% = 0.83$. The Liquid-heavy mix wins because this ratio measures return against 0%, not against Liquid.

1c Limits every optimiser must obey

To stop the optimiser putting everything in one asset, each weight is kept between 10% and 70%, and the weights must add up to 100%. The best weights are found by a standard solver (SLSQP) that tries weights until the goal cannot be improved.

$$0.10 \leq w_i \leq 0.70 \quad \sum_i w_i = 1$$

0.10, 0.70 lowest / highest allowed share of any asset

Example: Liquid ends up at the 70% cap in every year. Without the cap, risk parity would put 96.2% in Liquid.

1d Re-learn every year (walk-forward)

Every 12 months (each February) the 10-year window moves forward one year and steps 1a-1c are repeated. The new target is used for the next 12 months.

$$w^{(Feb\ Y)} = \text{optimiser}(r_{Feb\ Y-10}, \dots, r_{Jan\ Y})$$

Y the year of the decision (2010, 2011, ..., 2019)

Example: Feb 2011 uses Feb 2001 – Jan 2011, ..., Feb 2019 uses Feb 2009 – Jan 2019. 10 decisions in total.

2 The instalment arrives

On the 1st of every month you pay in the same amount. The formula allows the amount to grow every year (a “step-up”), but in this project the growth rate is 0, so it is always ₹10,000.

$$C_t = A \times (1 + g)^{\lfloor t/12 \rfloor}$$

C_t the instalment paid in month t

t the month number, counting from 0 (Feb 2010 is $t = 0$, Dec 2019 is $t = 118$)

A the starting instalment = ₹10,000

g yearly step-up of the instalment = 0 (no step-up)

$\lfloor t/12 \rfloor$ number of full years passed ($t \div 12$, rounded down)

Example: Feb 2010: $C = 10,000 \times (1 + 0)^0 = \text{₹10,000}$. Because $g = 0$, every one of the 119 months is ₹10,000, so the total paid in is $119 \times \text{₹10,000} = \text{₹1,190,000}$.

3 Split the instalment (smart)

Instead of copying the recipe, the ₹10,000 is sent *first* to the jars that are below their target. Think of topping up the emptiest glasses first. This pulls the portfolio back towards the target **without selling anything**.

$$V = \sum_i h_i + C_t \quad \text{gap}_i = \max(w_i V - h_i, 0) \quad G = \sum_i \text{gap}_i$$

$$\text{If } G \geq C_t : \quad x_i = \text{gap}_i \times \frac{C_t}{G}$$

$$\text{If } G < C_t : \quad x_i = \text{gap}_i + w_i \times (C_t - G)$$

h_i rupees already in jar i before buying

V total wealth after this month's instalment

$w_i V$ what jar i *should* hold

gap_i how many rupees jar i is short of its target (0 if it is already above)

G all the gaps added together

x_i rupees that go into jar i this month

Example: Feb 2010: jars hold Nifty ₹0.00, Gold ₹0.00, Liquid ₹0.00; $V = ₹0.00 + ₹10,000 = ₹10,000.00$.

Nifty: $\max(10.00\% \times ₹10,000.00 - ₹0.00, 0) = \max(₹1,000.00 - ₹0.00, 0) = ₹1,000.00$

Gold: $\max(20.00\% \times ₹10,000.00 - ₹0.00, 0) = \max(₹2,000.00 - ₹0.00, 0) = ₹2,000.00$

Liquid: $\max(70.00\% \times ₹10,000.00 - ₹0.00, 0) = \max(₹7,000.00 - ₹0.00, 0) = ₹7,000.00$

$G = ₹10,000.00$: all three jars were below target, so the gaps add up to exactly the ₹10,000 available and each jar gets exactly its gap → Nifty ₹1,000.00, Gold ₹2,000.00, Liquid ₹7,000.00. (In the very first month all jars are empty, so the gaps are exactly $w \times ₹10,000$.)

4 Buy: put the money in the jars

Each piece of the instalment is added to its jar.

$$h_i \leftarrow h_i + x_i$$

h_i rupees in jar i (Nifty, Gold or Liquid)

$\leftarrow$ "becomes": the new value replaces the old one

x_i rupees bought for jar i this month

Example: Feb 2010: Nifty: ₹0.00 + ₹1,000.00 = ₹1,000.00, Gold: ₹0.00 + ₹2,000.00 = ₹2,000.00, Liquid: ₹0.00 + ₹7,000.00 = ₹7,000.00.

5 Rebalance? Only if something drifts more than 5% (band rule)

After buying, the engine measures how far each jar's share is from its target. If the biggest gap (the *drift*) is more than 5 percentage points, the jars are reset to the target exactly as in the calendar rule. Otherwise nothing is sold.

$$drift = \max_i \left| \frac{h_i}{\sum_j h_j} - w_i \right| \quad \text{rebalance if } drift > 0.05$$

$$trade_i = w_i \times \sum_j h_j - h_i$$

$h_i / \sum_j h_j$ jar i's actual share of the total

$|\dots|$ size of the gap, ignoring its sign

$\max_i$ the biggest of the three gaps

0.05 the 5% band

Example: Feb 2011: Nifty $|10.00\% - 10.00\%| = 0.00\%$, Gold $|20.00\% - 20.00\%| = 0.00\%$, Liquid $|70.00\% - 70.00\%| = 0.00\%$. Drift = 0.00%, below 5% → **no rebalance**. The smart split already filled the gaps. Over 10 years the band was never crossed (0 rebalances), so this strategy never sold anything.

6 Pay the trading fee

Every rupee bought or sold costs 0.1% (10 paise per ₹100), like a broker's charge. The fee is taken from all jars in proportion to their size, so the split does not change.

$$Fee = c \times \left(\sum_i buys_i + \sum_i sells_i \right)$$

$$h_i \leftarrow h_i \times \left(1 - \frac{Fee}{V_{start}} \right)$$

c cost rate = 0.1% = 0.001

$buys_i, sells_i$ rupees bought / sold in jar i this month

V_{start} total of all jars just before the fee

$1 - Fee/V_{start}$ the fraction of each jar that is kept

Example: Feb 2010: traded ₹10,000.00 (only the instalment was bought). $Fee = 0.001 \times ₹10,000.00 = ₹10.00$. Each jar keeps $1 - ₹10.00 \div ₹10,000.00 = 0.999000$ of its value → Nifty ₹999.00, Gold ₹1,998.00, Liquid ₹6,993.00.

7 The market moves

During the month each jar grows or shrinks by that asset's return. A +1% month turns ₹100 into ₹101; a −2% month turns it into ₹98.

$$h_i \leftarrow h_i \times (1 + r_{i,t})$$

$r_{i,t}$ the return of asset i in month t (from 00_raw_data/monthly_returns.csv)

Example: Feb 2010 returns: Nifty +0.93%, Gold +3.17%, Liquid +0.31%.

Nifty: ₹999.00 × (1 + 0.0093) = ₹1,008.32, Gold: ₹1,998.00 × (1 + 0.0317) = ₹2,061.36, Liquid: ₹6,993.00 × (1 + 0.0031) = ₹7,015.03. End-of-month total **₹10,084.71**.

8 Measure this month's return (TWR)

How well did the *strategy* do this month, ignoring the fact that you added new money? Compare the value at the end with the value right after the money went in.

$$TWR_t = \frac{V_{end}}{V_{start}} - 1$$

V_{end} total of all jars at the end of the month

V_{start} total just after buying (and any rebalancing), before the fee

TWR_t time-weighted return of month t (the fee counts as part of the loss)

Example: Feb 2010: ₹10,084.71 ÷ ₹10,000.00 − 1 = **+0.85%**.

🔄 **Repeat steps 2-8 for every month**, Feb 2010 → Dec 2019 (119 times). Each month starts with last month's end-of-month jars. After the last month the jars hold **₹1,770,116**.

The smart split in a later month (the jars are no longer empty)

3 Split the instalment (smart)

Instead of copying the recipe, the ₹10,000 is sent *first* to the jars that are below their target. Think of topping up the emptiest glasses first. This pulls the portfolio back towards the target **without selling anything**.

$$V = \sum_i h_i + C_t \quad \text{gap}_i = \max(w_i V - h_i, 0) \quad G = \sum_i \text{gap}_i$$

$$\text{If } G \geq C_t : \quad x_i = \text{gap}_i \times \frac{C_t}{G}$$

$$\text{If } G < C_t : \quad x_i = \text{gap}_i + w_i \times (C_t - G)$$

h_i rupees already in jar i before buying

V total wealth after this month's instalment

$w_i V$ what jar i *should* hold

gap_i how many rupees jar i is short of its target (0 if it is already above)

G all the gaps added together

x_i rupees that go into jar i this month

Example: Feb 2011: jars hold Nifty ₹11,422.46, Gold ₹24,464.87, Liquid ₹89,521.56; $V = ₹125,408.90 + ₹10,000 = ₹135,408.90$.

Nifty: $\max(10.00\% \times ₹135,408.90 - ₹11,422.46, 0) = \max(₹13,540.89 - ₹11,422.46, 0) = ₹2,118.43$

Gold: $\max(20.00\% \times ₹135,408.90 - ₹24,464.87, 0) = \max(₹27,081.78 - ₹24,464.87, 0) = ₹2,616.91$

Liquid: $\max(70.00\% \times ₹135,408.90 - ₹89,521.56, 0) = \max(₹94,786.23 - ₹89,521.56, 0) =$

₹5,264.66

$G = ₹10,000.00$: all three jars were below target, so the gaps add up to exactly the ₹10,000 available and each jar gets exactly its gap → Nifty ₹2,118.43, Gold ₹2,616.91, Liquid ₹5,264.66.

Money example: Feb 2011, all steps together

February 2011: the optimiser has just re-learned the target from Feb 2001 – Jan 2011 and the smart split fills the gaps.

	Nifty	Gold	Liquid	Total
① Start of month (last month's end)	₹11,422.46	₹24,464.87	₹89,521.56	₹125,408.90
② Instalment ₹10,000 split into	₹2,118.43	₹2,616.91	₹5,264.66	₹10,000.00
③ Jars after buying	₹13,540.89	₹27,081.78	₹94,786.23	₹135,408.90
③ Share of total after buying	10.0%	20.0%	70.0%	100%
Target split w	10.0%	20.0%	70.0%	100%
④ Rebalance?	No			
⑤ Fee (0.1% of ₹10,000.00 traded)	taken from every jar in proportion			–₹10.00
⑤ Jars after fee	₹13,539.89	₹27,079.78	₹94,779.23	₹135,398.90
⑥ Market return this month r	–3.03%	+3.95%	+0.55%	
⑥ End of month	₹13,129.98	₹28,148.76	₹95,301.24	₹136,579.98

The new money went mostly where it was needed, so the split after buying equals the target and no selling was necessary. In the market move Nifty fell 3.03% while Gold rose 3.95%. The month's return: $₹136,579.98 \div ₹135,408.90 - 1 = +0.86\%$. Paid in so far: ₹130,000.

Scoring after 119 months

9a Score 1: yearly return on your money (XIRR)

Your instalments went in at different times, so a simple average is wrong. XIRR is the single yearly interest rate that, if a bank paid it on every instalment from the day you paid it, would give exactly your final value.

$$\sum_{k=0}^{118} C_k \times (1 + XIRR)^{(119-k)/12} = V_{final}$$

C_k instalment paid at the start of month k (₹10,000)

$(119 - k)/12$ years that instalment was invested until the end

V_{final} the value at the end of Dec 2019

Example: 119 instalments of ₹10,000 grew to **₹1,770,116**. The rate that makes both sides equal (found by trial and error, brentq) is **XIRR = 7.77%** a year. (For comparison, a 100% Liquid SIP gives 7.28%.)

9b Score 2: how bumpy is the ride (volatility)

Volatility measures how much the monthly return jumps around its average. Bigger jumps mean a scarier ride.

$$\sigma = \text{std}(TWR_1, \dots, TWR_{119}) \times \sqrt{12}$$

std standard deviation: the typical distance from the average

$\sqrt{12}$ turns a monthly figure into a yearly one

Example: monthly std = 0.93% $\rightarrow \sigma = 0.93\% \times 3.464 = 3.22\%$ a year.

9c Score 3: reward per unit of risk (Sharpe vs Liquid)

Liquid (a T-bill fund) earns interest with almost no risk. So the fair question is: how much *extra* return did the strategy earn above Liquid, per unit of bumpiness?

$$Sharpe_{vs\ Liquid} = \frac{12 \times \text{mean}(TWR_t - r_{Liquid,t})}{\sigma}$$

$r_{Liquid,t}$ Liquid's return in month t ($\approx$ the 91-day T-bill rate)

$12 \times \text{mean}(\dots)$ average extra return per year

σ volatility from score 2

Example: average extra return = 0.064% a month $\rightarrow \times 12 = 0.76\%$ a year; $0.76\% \div 3.22\% = 0.24$.
(Measured against 0% instead of Liquid it would be 2.49.)

9d Score 4: worst fall (maximum drawdown)

The biggest drop from a high point to a later low point, as if ₹1 had followed the strategy's monthly returns. It answers: "what is the worst loss I would have felt?"

$$G_t = \prod_{k \leq t} (1 + TWR_k) \quad MDD = \min_t \left(\frac{G_t}{\max_{k \leq t} G_k} - 1 \right)$$

G_t growth of ₹1 up to month t

$\max_{k \leq t} G_k$ the highest point reached so far

MDD the deepest fall below a previous high

Example: ₹1 grew to ₹1.2368 by Nov 2011, then fell to ₹1.2151 by Dec 2011: $1.2151 \div 1.2368 - 1 = -1.8\%$.

Paid in

₹1,190,000

Final value

₹1,770,116

XIRR

7.8%

Volatility

3.2%

Sharpe vs Liquid

0.24

Max drawdown

-1.8%

Strategy 6 Optimized SIP (average of 4 and 5)

The project's proposed strategy: every year take the average of the risk-parity and max-Sharpe splits, send each instalment to the jars that are below target, and sell only if something drifts more than 5%.

Setting	This strategy
Where the target split w comes from	$\frac{1}{2}$ risk parity + $\frac{1}{2}$ max Sharpe (re-fit yearly)
How each instalment is split	Smart (fill the gaps first)
When the whole portfolio is rebalanced	Only if drift > 5% (band)
Instalment, fee, market move, TWR, scoring	Same for all six strategies

The steps and their formulas

1a Look back 10 years: average return μ and covariance Σ

Before each yearly decision the optimiser studies only the **past** 120 months. It never sees the future. For the first SIP month it uses Feb 2000 – Jan 2010.

$$\mu_i = \frac{1}{120} \sum_{k=1}^{120} r_{i,t-k} \quad \Sigma_{ij} = \frac{1}{119} \sum_{k=1}^{120} (r_{i,t-k} - \mu_i)(r_{j,t-k} - \mu_j)$$

μ_i average monthly return of asset i over the last 120 months

Σ_{ij} covariance: how assets i and j move together (Σ_{ii} is asset i 's variance = volatility²)

$r_{i,t-k}$ return of asset i , k months ago

Example: Feb 2000 – Jan 2010 (as yearly figures): average return Nifty 16.8%, Gold 15.3%, Liquid 6.2%; volatility Nifty 28.0%, Gold 16.8%, Liquid 0.5%. Liquid barely moves (volatility 0.5%): it looks almost risk-free to the optimiser.

1b Run both optimisers, then average them

The Optimized SIP asks both experts from Strategies 4 and 5 and takes the middle of their answers. Averaging makes the target steadier than either on its own.

$$w^{RP} = \arg \min_w \sum_i (RC_i - \overline{RC})^2 \quad w^{MS} = \arg \max_w \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}}$$

$$w_i^{Opt} = \frac{1}{2} w_i^{RP} + \frac{1}{2} w_i^{MS}$$

w^{RP} risk-parity weights (Strategy 4, equal risk contributions)

w^{MS} max-Sharpe weights (Strategy 5, best return per risk)

w^{Opt} the Optimized SIP target

Example: Feb 2010: Nifty: $\frac{1}{2} \times 10.66\% + \frac{1}{2} \times 10.00\% = \mathbf{10.33\%}$, Gold: $\frac{1}{2} \times 19.34\% + \frac{1}{2} \times 20.00\% = \mathbf{19.67\%}$, Liquid: $\frac{1}{2} \times 70.00\% + \frac{1}{2} \times 70.00\% = \mathbf{70.00\%}$.

1c Limits every optimiser must obey

To stop the optimiser putting everything in one asset, each weight is kept between 10% and 70%, and the weights must add up to 100%. The best weights are found by a standard solver (SLSQP) that tries weights until the goal cannot be improved.

$$0.10 \leq w_i \leq 0.70 \quad \sum_i w_i = 1$$

0.10, 0.70 lowest / highest allowed share of any asset

Example: Liquid ends up at the 70% cap in every year. Without the cap, risk parity would put 96.2% in Liquid.

1d Re-learn every year (walk-forward)

Every 12 months (each February) the 10-year window moves forward one year and steps 1a-1c are repeated. The new target is used for the next 12 months.

$$w^{(Feb\ Y)} = \text{optimiser}(r_{Feb\ Y-10}, \dots, r_{Jan\ Y})$$

Y the year of the decision (2010, 2011, ..., 2019)

Example: Feb 2011 uses Feb 2001 – Jan 2011, ..., Feb 2019 uses Feb 2009 – Jan 2019. 10 decisions in total.

2 The instalment arrives

On the 1st of every month you pay in the same amount. The formula allows the amount to grow every year (a “step-up”), but in this project the growth rate is 0, so it is always ₹10,000.

$$C_t = A \times (1 + g)^{\lfloor t/12 \rfloor}$$

C_t the instalment paid in month t

t the month number, counting from 0 (Feb 2010 is $t = 0$, Dec 2019 is $t = 118$)

A the starting instalment = ₹10,000

g yearly step-up of the instalment = 0 (no step-up)

$\lfloor t/12 \rfloor$ number of full years passed ($t \div 12$, rounded down)

Example: Feb 2010: $C = 10,000 \times (1 + 0)^0 = \text{₹10,000}$. Because $g = 0$, every one of the 119 months is ₹10,000, so the total paid in is $119 \times \text{₹10,000} = \text{₹1,190,000}$.

3 Split the instalment (smart)

Instead of copying the recipe, the ₹10,000 is sent *first* to the jars that are below their target. Think of topping up the emptiest glasses first. This pulls the portfolio back towards the target **without selling anything**.

$$V = \sum_i h_i + C_t \quad \text{gap}_i = \max(w_i V - h_i, 0) \quad G = \sum_i \text{gap}_i$$

$$\text{If } G \geq C_t : \quad x_i = \text{gap}_i \times \frac{C_t}{G}$$

$$\text{If } G < C_t : \quad x_i = \text{gap}_i + w_i \times (C_t - G)$$

h_i rupees already in jar i before buying

V total wealth after this month's instalment

$w_i V$ what jar i *should* hold

gap_i how many rupees jar i is short of its target (0 if it is already above)

G all the gaps added together

x_i rupees that go into jar i this month

Example: Feb 2010: jars hold Nifty ₹0.00, Gold ₹0.00, Liquid ₹0.00; $V = ₹0.00 + ₹10,000 = ₹10,000.00$.

Nifty: $\max(10.33\% \times ₹10,000.00 - ₹0.00, 0) = \max(₹1,033.06 - ₹0.00, 0) = ₹1,033.06$

Gold: $\max(19.67\% \times ₹10,000.00 - ₹0.00, 0) = \max(₹1,966.94 - ₹0.00, 0) = ₹1,966.94$

Liquid: $\max(70.00\% \times ₹10,000.00 - ₹0.00, 0) = \max(₹7,000.00 - ₹0.00, 0) = ₹7,000.00$

$G = ₹10,000.00$: all three jars were below target, so the gaps add up to exactly the ₹10,000 available and each jar gets exactly its gap → Nifty ₹1,033.06, Gold ₹1,966.94, Liquid ₹7,000.00. (In the very first month all jars are empty, so the gaps are exactly $w \times ₹10,000$.)

4 Buy: put the money in the jars

Each piece of the instalment is added to its jar.

$$h_i \leftarrow h_i + x_i$$

h_i rupees in jar i (Nifty, Gold or Liquid)

$\leftarrow$ "becomes": the new value replaces the old one

x_i rupees bought for jar i this month

Example: Feb 2010: Nifty: ₹0.00 + ₹1,033.06 = ₹1,033.06, Gold: ₹0.00 + ₹1,966.94 = ₹1,966.94, Liquid: ₹0.00 + ₹7,000.00 = ₹7,000.00.

5 Rebalance? Only if something drifts more than 5% (band rule)

After buying, the engine measures how far each jar's share is from its target. If the biggest gap (the *drift*) is more than 5 percentage points, the jars are reset to the target exactly as in the calendar rule. Otherwise nothing is sold.

$$drift = \max_i \left| \frac{h_i}{\sum_j h_j} - w_i \right| \quad \text{rebalance if } drift > 0.05$$

$$trade_i = w_i \times \sum_j h_j - h_i$$

$h_i / \sum_j h_j$ jar i's actual share of the total

$|\dots|$ size of the gap, ignoring its sign

$\max_i$ the biggest of the three gaps

0.05 the 5% band

Example: Feb 2011: Nifty $|10.51\% - 10.51\%| = 0.00\%$, Gold $|19.49\% - 19.49\%| = 0.00\%$, Liquid $|70.00\% - 70.00\%| = 0.00\%$. Drift = 0.00%, below 5% → **no rebalance**. The smart split already filled the gaps. Over 10 years the band was never crossed (0 rebalances), so this strategy never sold anything.

6 Pay the trading fee

Every rupee bought or sold costs 0.1% (10 paise per ₹100), like a broker's charge. The fee is taken from all jars in proportion to their size, so the split does not change.

$$Fee = c \times \left(\sum_i buys_i + \sum_i sells_i \right)$$

$$h_i \leftarrow h_i \times \left(1 - \frac{Fee}{V_{start}} \right)$$

c cost rate = 0.1% = 0.001

$buys_i, sells_i$ rupees bought / sold in jar i this month

V_{start} total of all jars just before the fee

$1 - Fee/V_{start}$ the fraction of each jar that is kept

Example: Feb 2010: traded ₹10,000.00 (only the instalment was bought). $Fee = 0.001 \times ₹10,000.00 = ₹10.00$. Each jar keeps $1 - ₹10.00 \div ₹10,000.00 = 0.999000$ of its value → Nifty ₹1,032.03, Gold ₹1,964.97, Liquid ₹6,993.00.

7 The market moves

During the month each jar grows or shrinks by that asset's return. A +1% month turns ₹100 into ₹101; a −2% month turns it into ₹98.

$$h_i \leftarrow h_i \times (1 + r_{i,t})$$

$r_{i,t}$ the return of asset i in month t (from 00_raw_data/monthly_returns.csv)

Example: Feb 2010 returns: Nifty +0.93%, Gold +3.17%, Liquid +0.31%.

Nifty: ₹1,032.03 × (1 + 0.0093) = ₹1,041.65, Gold: ₹1,964.97 × (1 + 0.0317) = ₹2,027.29, Liquid: ₹6,993.00 × (1 + 0.0031) = ₹7,015.03. End-of-month total **₹10,083.97**.

8 Measure this month's return (TWR)

How well did the *strategy* do this month, ignoring the fact that you added new money? Compare the value at the end with the value right after the money went in.

$$TWR_t = \frac{V_{end}}{V_{start}} - 1$$

V_{end} total of all jars at the end of the month

V_{start} total just after buying (and any rebalancing), before the fee

TWR_t time-weighted return of month t (the fee counts as part of the loss)

Example: Feb 2010: ₹10,083.97 ÷ ₹10,000.00 − 1 = **+0.84%**.

🔄 **Repeat steps 2-8 for every month**, Feb 2010 → Dec 2019 (119 times). Each month starts with last month's end-of-month jars. After the last month the jars hold **₹1,774,231**.

The smart split in a later month (the jars are no longer empty)

3 Split the instalment (smart)

Instead of copying the recipe, the ₹10,000 is sent *first* to the jars that are below their target. Think of topping up the emptiest glasses first. This pulls the portfolio back towards the target **without selling anything**.

$$V = \sum_i h_i + C_t \quad \text{gap}_i = \max(w_i V - h_i, 0) \quad G = \sum_i \text{gap}_i$$

$$\begin{aligned} \text{If } G \geq C_t : \quad x_i &= \text{gap}_i \times \frac{C_t}{G} \\ \text{If } G < C_t : \quad x_i &= \text{gap}_i + w_i \times (C_t - G) \end{aligned}$$

h_i rupees already in jar i before buying

V total wealth after this month's instalment

$w_i V$ what jar i *should* hold

gap_i how many rupees jar i is short of its target (0 if it is already above)

G all the gaps added together

x_i rupees that go into jar i this month

Example: Feb 2011: jars hold Nifty ₹11,800.23, Gold ₹24,060.79, Liquid ₹89,522.71; $V = ₹125,383.73 + ₹10,000 = ₹135,383.73$.

Nifty: $\max(10.51\% \times ₹135,383.73 - ₹11,800.23, 0) = \max(₹14,225.85 - ₹11,800.23, 0) = ₹2,425.63$

Gold: $\max(19.49\% \times ₹135,383.73 - ₹24,060.79, 0) = \max(₹26,389.27 - ₹24,060.79, 0) = ₹2,328.47$

Liquid: $\max(70.00\% \times ₹135,383.73 - ₹89,522.71, 0) = \max(₹94,768.61 - ₹89,522.71, 0) =$

₹5,245.90

$G = ₹10,000.00$: all three jars were below target, so the gaps add up to exactly the ₹10,000 available and each jar gets exactly its gap → Nifty ₹2,425.63, Gold ₹2,328.47, Liquid ₹5,245.90.

Money example: Feb 2011, all steps together

February 2011: the optimiser has just re-learned the target from Feb 2001 – Jan 2011 and the smart split fills the gaps.

	Nifty	Gold	Liquid	Total
① Start of month (last month's end)	₹11,800.23	₹24,060.79	₹89,522.71	₹125,383.73
② Instalment ₹10,000 split into	₹2,425.63	₹2,328.47	₹5,245.90	₹10,000.00
③ Jars after buying	₹14,225.85	₹26,389.27	₹94,768.61	₹135,383.73
③ Share of total after buying	10.5%	19.5%	70.0%	100%
Target split w	10.5%	19.5%	70.0%	100%
④ Rebalance?	No			
⑤ Fee (0.1% of ₹10,000.00 traded)	taken from every jar in proportion			–₹10.00
⑤ Jars after fee	₹14,224.80	₹26,387.32	₹94,761.61	₹135,373.73
⑥ Market return this month r	–3.03%	+3.95%	+0.55%	
⑥ End of month	₹13,794.16	₹27,428.96	₹95,283.52	₹136,506.64

The new money went mostly where it was needed, so the split after buying equals the target and no selling was necessary. In the market move Nifty fell 3.03% while Gold rose 3.95%. The month's return: $₹136,506.64 \div ₹135,383.73 - 1 = +0.83\%$. Paid in so far: ₹130,000.

Scoring after 119 months

9a Score 1: yearly return on your money (XIRR)

Your instalments went in at different times, so a simple average is wrong. XIRR is the single yearly interest rate that, if a bank paid it on every instalment from the day you paid it, would give exactly your final value.

$$\sum_{k=0}^{118} C_k \times (1 + XIRR)^{(119-k)/12} = V_{final}$$

C_k instalment paid at the start of month k (₹10,000)

$(119 - k)/12$ years that instalment was invested until the end

V_{final} the value at the end of Dec 2019

Example: 119 instalments of ₹10,000 grew to **₹1,774,231**. The rate that makes both sides equal (found by trial and error, brentq) is **XIRR = 7.82%** a year. (For comparison, a 100% Liquid SIP gives 7.28%.)

9b Score 2: how bumpy is the ride (volatility)

Volatility measures how much the monthly return jumps around its average. Bigger jumps mean a scarier ride.

$$\sigma = \text{std}(TWR_1, \dots, TWR_{119}) \times \sqrt{12}$$

std standard deviation: the typical distance from the average

$\sqrt{12}$ turns a monthly figure into a yearly one

Example: monthly std = 0.91% → $\sigma = 0.91\% \times 3.464 = 3.17\%$ a year.

9c Score 3: reward per unit of risk (Sharpe vs Liquid)

Liquid (a T-bill fund) earns interest with almost no risk. So the fair question is: how much *extra* return did the strategy earn above Liquid, per unit of bumpiness?

$$Sharpe_{vs\ Liquid} = \frac{12 \times \text{mean}(TWR_t - r_{Liquid,t})}{\sigma}$$

$r_{Liquid,t}$ Liquid's return in month t ($\approx$ the 91-day T-bill rate)

$12 \times \text{mean}(\dots)$ average extra return per year

σ volatility from score 2

Example: average extra return = 0.066% a month → $\times 12 = 0.80\%$ a year; $0.80\% \div 3.17\% = 0.25$.
(Measured against 0% instead of Liquid it would be 2.54.)

9d Score 4: worst fall (maximum drawdown)

The biggest drop from a high point to a later low point, as if ₹1 had followed the strategy's monthly returns. It answers: "what is the worst loss I would have felt?"

$$G_t = \prod_{k \leq t} (1 + TWR_k) \quad MDD = \min_t \left(\frac{G_t}{\max_{k \leq t} G_k} - 1 \right)$$

G_t growth of ₹1 up to month t

$\max_{k \leq t} G_k$ the highest point reached so far

MDD the deepest fall below a previous high

Example: ₹1 grew to ₹1.2333 by Nov 2011, then fell to ₹1.2120 by Dec 2011: $1.2120 \div 1.2333 - 1 = -1.7\%$.

Paid in

₹1,190,000

Final value

₹1,774,231

XIRR

7.8%

Volatility

3.2%

Sharpe vs Liquid

0.25

Max drawdown

-1.7%

Summary All six side by side

Same data, same 119 months, same ₹10,000 instalments (₹1,190,000 paid in), same 0.1% fee.

Strategy	Final value	XIRR	Volatility	Sharpe vs Liquid	Max drawdown
1 · Equity (100% Nifty)	₹2,176,119	11.7%	15.3%	0.29	−23.7%
2 · Equal-weight $\frac{1}{3}$ each	₹1,850,460	8.6%	6.5%	0.28	−6.5%
3 · 60/20/20 annual	₹1,990,520	10.0%	8.8%	0.33	−8.9%
4 · Risk parity	₹1,778,154	7.9%	3.1%	0.26	−1.7%
5 · Max Sharpe	₹1,770,116	7.8%	3.2%	0.24	−1.8%
6 · Optimized ($\frac{1}{2} + \frac{1}{2}$)	₹1,774,231	7.8%	3.2%	0.25	−1.7%
Reference: 100% Liquid SIP		7.3%			

- **Most money:** Strategy 1, 100% Nifty (11.7% a year), but with the deepest fall (−23.7%).
- **Best reward per unit of risk, measured fairly against Liquid:** Strategy 3, 60/20/20 (Sharpe vs Liquid 0.33).
- **Smallest falls:** Strategies 4–6 (about −1.7%), because the optimisers put 70% in Liquid. They earned only about 0.5% a year more than a 100% Liquid SIP.

Flowchart

The whole process, formula at each step

DATA: monthly returns (00_raw_data)

$$r_{i,t} = P_{i,t}/P_{i,t-1} - 1$$

Nifty (+ dividends) · Gold in ₹ = \$ price × USD/INR · Liquid (T-bill index), Feb 2000 – Dec 2019

STEP 1: TARGET SPLIT w (each strategy's own rule)

1-3 · fixed
(1,0,0) · (1/3,1/3,1/3) ·
(0.6,0.2,0.2)

4 · risk parity

$$\min \sum (RC_i - \overline{RC})^2$$

5 · max Sharpe

$$\max w^\top \mu / \sqrt{w^\top \Sigma w}$$

6 · optimized

$$\frac{1}{2}w^{RP} + \frac{1}{2}w^{MS}$$

4-6: μ, Σ from the previous 120 months only · 10% ≤ w ≤ 70% · re-fit every February

MONTHLY SIP ENGINE · repeated 119 times (Feb 2010 → Dec 2019)

2 · Instalment

$$C_t = A(1 + g)^{\lfloor t/12 \rfloor}$$

A = ₹10,000, g = 0 → always ₹10,000

3 · Split the instalment

pro-rata (1-3)

$$x_i = w_i C_t$$

smart (4-6)

$$gap_i = \max(w_i V - h_i, 0)$$

fill gaps first, rest by w

4 · Buy

$$h_i \leftarrow h_i + x_i$$

5 · Rebalance?

none (1, 2)

calendar (3)

every 12 months

band (4-6)

$$\max_i |h_i / \sum h - w_i| > 5\%$$

$$\text{if yes: } h_i \leftarrow w_i \sum_j h_j$$

6 · Fee

$$Fee = 0.001 \times (buys + sells) \quad h_i \leftarrow h_i (1 - Fee/V_{start})$$

7 · Market move

$$h_i \leftarrow h_i (1 + r_{i,t})$$

8 · Month's return

$$TWR_t = V_{end}/V_{start} - 1$$

↻ next month: t → t + 1

STEP 9: SCORE EACH STRATEGY

$$\sum_k C_k (1 + XIRR)^{(119-k)/12} = V_{final}$$

$$\sigma = \text{std}(TWR) \sqrt{12}$$

$$Sharpe_{Liq} = 12 \text{ mean}(TWR - r_{Liq}) / \sigma$$

$$MDD = \min(G_t / \max G - 1)$$

COMPARE ALL SIX (07_comparison)

same data, months, instalments and fees → final value, XIRR, volatility, Sharpe vs Liquid, max drawdown